LOCAL QUASI-FINITENESS OF A LOCALLY FINITE TYPE MORPHISM OF ALGEBRAIC SPACES IS RELATIVE DIMENSION ZERO EVERYWHERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a direct reduction to the scheme case through a presentation; the two translation steps are stated in the span and rest on the chapter's definitions of the two notions for algebraic spaces, not on an omitted argument.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a morphism of algebraic spaces to a diagram of schemes. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 7086-7095. Produces the presentation h : U -> V.
- Local quasi-finiteness and relative dimension for algebraic spaces are etale local on the source and target. used at proof lines 7096-7099. Transfers both conditions between f and h.
- The scheme case: local quasi-finiteness equals relative dimension zero. Morphisms, Lemma morphisms-lemma-locally-quasi-finite-rel-dimension-0, used at proof lines 7100-7101. Is the input applied to h.

Source reference. Original lemma 0397: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4262–4262. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type. Then f is locally quasi-finite if and only if f has relative dimension 0 at each $x \in |X|$.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes and the vertical arrows are surjective and étale, see Spaces, Lemma 02X1. The definitions imply that h is locally quasi-finite if and only if f is locally quasi-finite, and that f has relative dimension 0 at all $x \in |X|$ if and only if h has relative dimension 0 at all $u \in U$. Hence the result follows from the result for h which is Morphisms, Lemma 0397. $\square$